

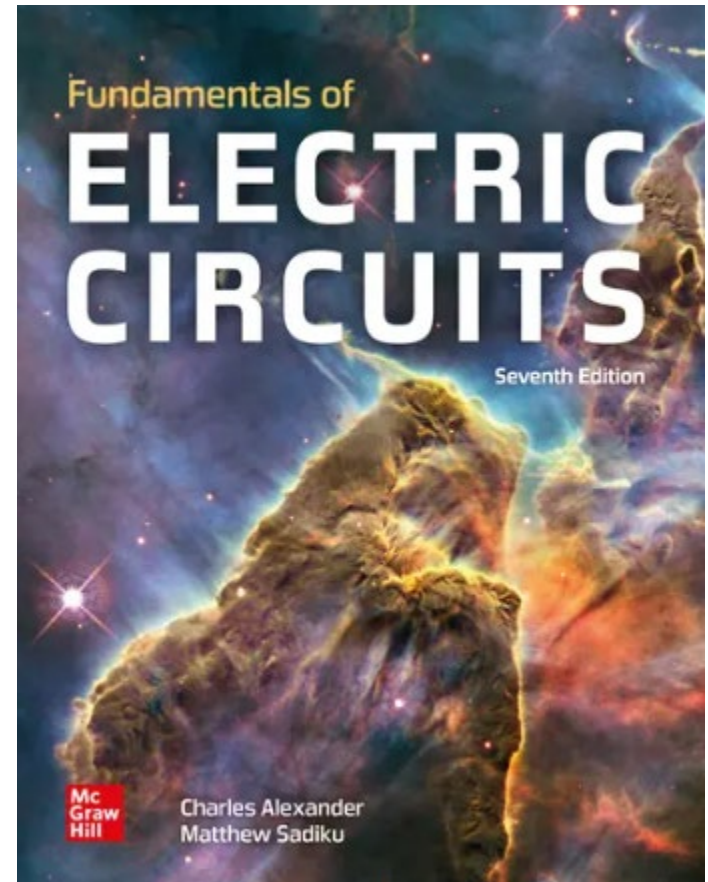
# Electrical Science

## ENGR 2303

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# Course Materials

**Textbook:** “Fundamentals of Electric Circuits”,  
by Alexander and Sadiku.



**Course Website:**

<http://www.engineering.uco.edu/~mbingabr>

# Course Topics

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- Ohm's Law, Kirchhoff's Law Series and Parallel Resistor
- Nodal and Loop Analysis Techniques
- Operational Amplifiers, Op-Amp Models
  - Test 1
- Superposition, Thevenin's and Norton's Theorem
- Maximum Power Transfer
- Capacitor and Inductor RLC Circuit
- RC Operational Amplifier Circuits
  - Test 2
- First Order Circuit
- AC Steady-State Analysis
- Instantaneous and Average Power
- Variable-Frequency Network Analysis and Performance (Filters)
  - Final

# **Circuit Theory**

## **Chapter 1**

### **Basic Concepts**

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# Basic Concepts - Chapter 1

1.1 Systems of Units.

1.2 Electric Charge.

1.3 Current.

1.4 Voltage.

1.5 Power and Energy.

1.6 Circuit Elements.

# 1.1 System of Units

**The derived units commonly used in electric circuit theory**

| Quantity              | Unit    | Symbol   |
|-----------------------|---------|----------|
| electric charge       | coulomb | C        |
| electric potential    | volt    | V        |
| resistance            | ohm     | $\Omega$ |
| conductance           | siemens | S        |
| inductance            | henry   | H        |
| capacitance           | farad   | F        |
| frequency             | hertz   | Hz       |
| force                 | newton  | N        |
| energy, work          | joule   | J        |
| power                 | watt    | W        |
| magnetic flux         | weber   | Wb       |
| magnetic flux density | tesla   | T        |

| Factor     | Prefix | Symbol |
|------------|--------|--------|
| $10^9$     | giga   | G      |
| $10^6$     | mega   | M      |
| $10^3$     | kilo   | k      |
| $10^{-2}$  | centi  | c      |
| $10^{-3}$  | milli  | m      |
| $10^{-6}$  | micro  | $\mu$  |
| $10^{-9}$  | nano   | n      |
| $10^{-12}$ | pico   | p      |

**Decimal multiples and  
submultiples of SI units**

# 1.2 Electric Charges

- **Charge** is an electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**.
- The charge **e** on one electron is negative and equal in magnitude to  **$1.602 \times 10^{-19}$  C** which is called as electronic charge. The charges that occur in nature are **integral multiples** of the electronic charge.

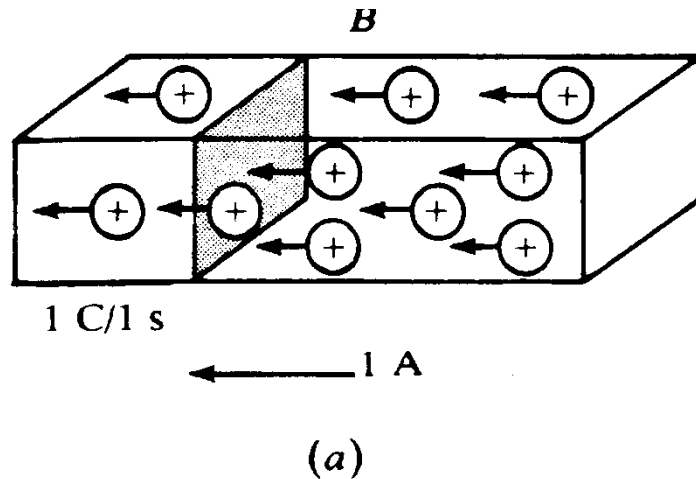
# 1.3 Current (1)

- Electric current  $i = dq/dt$ . The unit of ampere can be derived as  $1 \text{ A} = 1 \text{ C/s}$ .
- A **direct current (dc)** is a current that remains constant with time.
- An **alternating current (ac)** is a current that varies sinusoidally with time.  
(reverse direction)

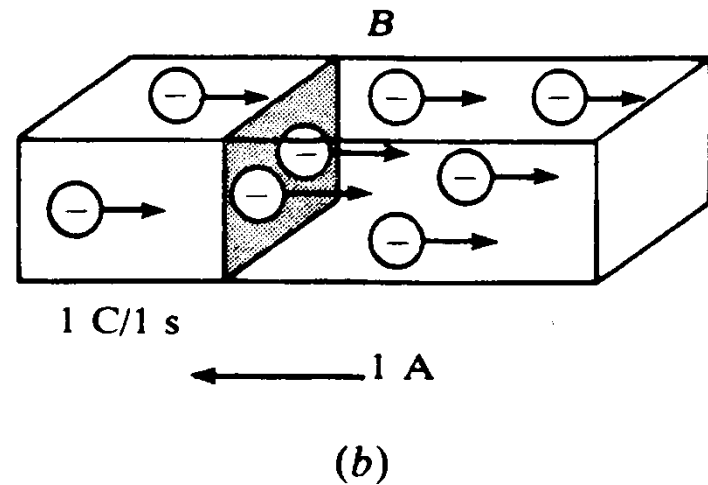


# 1.3 Current (2)

- The direction of current flow



Positive ions



Negative ions

## 1.3 Current (3)

### Example 1

A conductor has a constant current of 5 A.

How many electrons pass a fixed point on the conductor in one minute?

## 1.3 Current (4)

### Solution

Total no. of charges pass in 1 min is given by  
 $5 \text{ A} = (5 \text{ C/s})(60 \text{ s/min}) = 300 \text{ C/min}$

Total no. of electronics pass in 1 min is given

$$\begin{aligned}\text{Number of electrons} &= 300 / (1.602 \times 10^{-19}) \\ &= 187.3 \times 10^{19}\end{aligned}$$

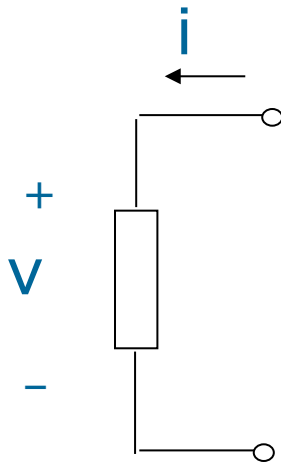
# 1.4 Voltage (1)

- Voltage (or potential difference) is the **energy** required to move a **unit charge** through an element, measured in volts (V).
- Mathematically,  $v_{ab} = dw / dq$  (volt)
  - **w** is energy in joules (J) and **q** is charge in coulomb (C).
- Electric voltage,  $v_{ab}$ , is always **across the circuit element** or **between two points in a circuit**.
  - $v_{ab} > 0$  means the potential of **a** is higher than potential of **b**.
  - $v_{ab} < 0$  means the potential of **a** is lower than potential of **b**.

# 1.5 Power and Energy (1)

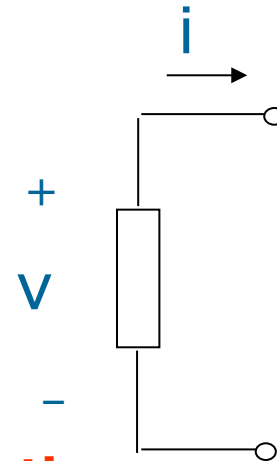
- Power is the time rate of expending or absorbing energy, measured in watts (W).

- Mathematical expression:  $p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$



**$P = +vi$**   
**absorbing power**

**Passive sign convention**



**$p = -vi$**   
**supplying power**

# 1.5 Power and Energy (2)

- The law of conservation of energy

$$\sum p = 0$$

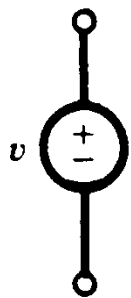
- **Energy** is the capacity to do work, measured in joules (J).

- Mathematical expression  $w = \int_{t_0}^t p dt = \int_{t_0}^t v i dt$

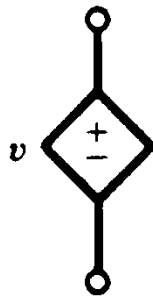
# 1.6 Circuit Elements (1)

## Active Elements

## Passive Elements



(a)



(b)



(c)



(d)



(e)



(f)



(g)

Independent sources      Dependent sources

- A dependent source is an active element in which the source quantity is controlled by another voltage or current.
- They have four different types: VCVS, CCVS, VCCS, CCCS. Keep in mind the signs of dependent sources.

# 1.6 Circuit Elements (2)

## Example 2

Obtain the voltage  $v$  in the branch shown in Figure 2.1.1P for  $i_2 = 1\text{A}$ .

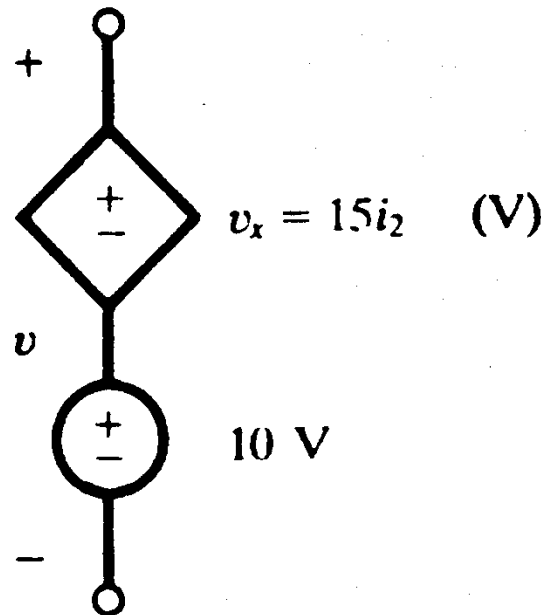


Figure 2.1.1P



# 1.6 Circuit Elements (3)

## Solution

Voltage  $v$  is the sum of the voltage-independent 10-V source and the voltage-dependent voltage source  $v_x$ .

Note that the factor 15 multiplying the control current carries the units  $\Omega$ .

Therefore,  $v = 10 + v_x = 10 + 15(1) = 25 \text{ V}$